Gautam Sharma

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B.Tech. - Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.73 (Current)	2021 - Present
B.Tech. Minor	Indian Institute of Technology, Guwahati	7.00 (Current)	2022 - Present
Senior Secondary	CBSE Board	95.20%	2021
Secondary	CBSE Board	95.80%	2019

Projects

- **Quantum Powered Business Rule Engine** *Dec 2023*
  Group Project, ***Gold Medal, 12th Inter IIT Tech Meet****, IIT Madras* bit.ly/3tCRDCf
  – Built a business rule engine using **Quantum Computing**, employing DWave's **dwave-system** which makes the process of re-accommodating passengers affected by flight delays/cancellation **easier and faster over classical methods**.
  – Solved the famous **Passenger Recovery Problem** by formulating a mixed integer programme.
  – For preprocessing, used **Prune and Search**, and **Multithreading**. Used **Dynamic Programming** to speed up the process.
  – Solved the problem classically using **gurobipy**, and as a **CQM**, and **BQM** using **Quantum Annealing**.
  – For post-processing, used **Flow Network Algorithms**, and displayed all the solutions using **Streamlit**.
- **Investing use of LLMs in verification of hardware design** *Nov 2023*
  Group Project, Guide: Dr. Chandan Karfa bit.ly/3O5bIbC
  – Tested the capabilities of GPT-4 in **generating independent assertions** about a hardware design from Google's OpenTitan.
  – Generated the assertions in **SVA** by giving verbal prompts and tested their validity in **EDA playground**.
  – Found out a bug in Open Titan's implementation of ROM Control, devised methods to prompt engineer for the above tasks.
- **Competitive Programming Helper Discord Bot** *Dec 2022*
  Group Project, Coding Club, IIT Guwahati bit.ly/3XBsRgl
  – Built a discord bot using **discord.py** which makes the process of planning CP practice effortless for a user.
  – Incorporated features such as: problem recommendation, track of rating changes, host duel between users, contest reminders, record all problems solved by a user, leaderboard, graphical statistics, support for Codeforces and AtCoder.
  – Used **Firebase** for storing user data, **Beautiful Soup** for web-scraping, and **Codeforces**, **Clist**, etc. API for API calls.
- **Quiz-Maker** *Jun 2023*
  Self Project, DIY Projects, Coding Club, IIT Guwahati bit.ly/446CPZT
  – Implemented a web-based application using **React.js** and **Node.js** which enables a user in creating quizzes on any topic.
  – Incorporated features such as: adding/modifying questions/quizzes, timed attempt of quizzes, and result of each attempt.
  – Used **MongoDB** for storing quizzes, user's past attempts, and credentials while used **antd** elements in the frontend. Used **bcrypt.js** for encryption and decryption and **Redux** for state management.
- **Database in C** *Jun 2023 - Ongoing*
  Self Project bit.ly/3CTskwS
  – Developed a program in **C** language, which attempts to replicate a full-fledged database management system.
  – Structured on **B+ trees**, the database supports insertion and selection operation to the database.
- **Eight puzzle solver** *Oct 2022*
  Guide: Prof. Pinaki Mitra | Course Project: System Software Lab bit.ly/46JbGO4
  – Developed a program in **Python** which solves the famous eight-puzzle problem deterministically.
  – Implemented the **A* algorithm** on the manhattan distances between the cells and assessed the reachability condition of the final state from the start state by comparing the parity of inversions in the start and final states.
- **Socket based Chat System** *Oct 2023*
  Guide: Prof. Moumita Patra | Course Project: Networks Lab bit.ly/3PXnXrV
  – Implemented a one server-multiple client program in C, supporting messages broadcast and one-to-one private chat.
  – Used **pthread.h** library for **multithreading** and **socket.h** library for creating **TCP sockets**.

Achievements

- **ICPC Mathura Prelims 2023**: Achieved **AIR 163** (**6th in IIT Guwahati**, all batches combined) among 1000+ teams.
- **Codeforces**: Achieved maximum rating of **1667 (Expert)** (Handle: *vimruLover*)
  – Codeforces Round 881: Secured **Global Rank 90** among 28,000+ participants. (Handle: *g-s01)*
  – Codeforces Round 874: Secured **Global Rank 236** among 30,600+ participants.
- **AtCoder**: Achieved maximum rating of **1102 (5 Kyu)** (Handle: *vimruLover*)
  – AtCoder Beginner Contest 308: Secured **Global Rank 634** among 10,000+ participants.
- **CodeChef**: Achieved maximum rating of **1968 (4 stars)** (Handle: *sharma_gautam*)
  – Starters 94: Secured **Global Rank 59** among 2,400+ participants.
- **12th Inter IIT Tech Meet**: Secured **Rank 1** among 100+ students from **20 IITs** in the **Quantum Computing Challenge**.
- **APAC Terminal 2023**:Secured **Rank 4** at **Asia-Pacific Terminal** by **Citadel Securities** and **Correlation One**.
- **Amazon ML Summer School 2023**:Selected to attend various AI/ML lectures by Amazon from students all over India.

- **ICPC-de-Tryst-2023**: Secured **Rank 1** at ICPC-de-Tryst event held at IIT Delhi.
- **Kriti-2023**: Secured **Rank 1** at Inter-Hostel competitive programming competition under Kriti'23 (Junior Section).
- **Abstractify-2023**: Secured **Rank 3** among participants from top 8 IITs in the Quant Winter Trading School, IIT Delhi.
- **Too Buggy**: Secured **Rank 2** (highest score, ties broken on seniority basis) in **Too Buggy** CP competition held at IIT Guwahati.
- **JEE Advanced 2021**: Secured **AIR 530** among 0.15 million students across the country.
- **JEE Mains 2021**: Secured **AIR 359** among 0.94 million students across the country.
- **KVPY SX 2021**: Secured **AIR 397** and the KVPY scholarship given to **top 2000** students across the country.
- **JSTSE 2018**: Secured **Rank 17 (out of 19,331)** and the JSTSE (now MVPP) scholarship given to **top 160** students in Delhi.

Technical Skills

- **Programming**: C, C++, Python*, JavaScript
- **Web Technologies**: React.js*, Node.js*, HTML, CSS
- **Optimization frameworks**: Gurobi, OR Tools
- **Database Management**: SQL, MongoDB, Firebase*
- **Quantum Computing**: Qiskit, DWave
- **Miscellaneous**: LaTeX, Figma, Markdown, Bash*

** Elementary proficiency*

Key courses taken

- **Computer Science**: Data Structures & Algorithms, Computer Architecture & Organization, Database Management Systems, Computer Networks, Operating Systems, Formal Methods for System Verification
- **Mathematics**: Linear Algebra, Number Theory, Probability & Random Processes, Discrete Mathematics

Positions of Responsibility

- **Competitive Programming Head**: Coding Club, IIT Guwahati *Apr 2023 - Ongoing*
  - **HeadStart**: Organized an introductory course on competitive programming, which had 1000+ signups.
  - **Coding Week-2023**: Organized a competitive programming contest which had 350+ registrations.
  - **Prepbit-2023**: Organized multiple contests to help everyone in preparing for the intern season, which had 300+ signups.
- **Coordinator, Competitive Programming Module**: Coding Club, IIT Guwahati *Aug 2022 - Mar 2023*
  - **CPP101**: Made a course on **C++**, which covered **C++ keywords**, **OOPS**, **Streams** and **Threads**, with 250+ registrations.
  - **Codebinge-2022**: Organised a Competitive Programming Contest at IITG which had 400+ registrations.
- **Leader, Live Class Coordination Team**: SOPAN-20, IIT Guwahati *Aug 2023 - Ongoing*
  - Mentoring and coordinating live classes among other logistics for 20 underprivileged students enrolled in SOPAN-20 initiative of IIT Guwahati, to help them prepare for JEE.
- **Mentor**: CSEA, IIT Guwahati *Nov 2022 - Ongoing*
  - Mentored eight first-years to help them adapt to their new college life while also making them familiar with the institute.
- **General Executive**: SAIL, IIT Guwahati *Aug 2022 - Nov 2022*
  - Responsible to connect with IITG Alumni for various workshops/talks/events for the benefit of the IITG junta/campus.